

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 19:00 UTC

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Generated 2026-05-12 19:00 UTC

- Cycles recorded: **671**
- Time span: 453.3h (~1.48 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	634
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	37
Schur-Weyl Duality	24
Representation Theory of Symmetric Groups	17
Matroid Theory	16
Free Probability	14
Additive Combinatorics	11
Free Probability Theory	11
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4

Field	Cycles
Algebraic Geometry of Secant Varieties	4
Persistent Homology	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-12 11:48 UTC	INCONCLUSIVE	Schur-Weyl Decomposition Component Count
2026-05-12 12:40 UTC	INCONCLUSIVE	vs. Monotone Circuit Siz
2026-05-12 13:19 UTC	INCONCLUSIVE	Real Radical Degree Lower Bound for AC ⁰ Circuits
2026-05-12 13:27 UTC	INCONCLUSIVE	Computing PARITY
2026-05-12 14:34 UTC	INCONCLUSIVE	Noncommutative Fourier Coefficient Spread and Read-Twice BP Size
2026-05-12 15:07 UTC	INCONCLUSIVE	Invariant Ring Generator Count Gap in Read-Twice BPs for IP ₂
2026-05-12 15:52 UTC	INCONCLUSIVE	Noncommutative Fourier Coefficient Sum Bounded by Log-Size for Re
2026-05-12 16:08 UTC	INCONCLUSIVE	Schur-Weyl Multiplicity Gap in Symmetric Powers of Permanent vs D
2026-05-12 16:21 UTC	INCONCLUSIVE	Additive Energy of Truth Table Bounds ACC ⁰ Circuit Size for Expli
2026-05-12 17:13 UTC	INCONCLUSIVE	Plethysm Multiplicity Inverse Proportional to Monotone Circuit Si
2026-05-12 17:24 UTC	INCONCLUSIVE	Ehrhart Polynomial Coefficient Sum Bounded by Resolution Proof Si
		Noncommutative L [∞] Norm Lower Bounds Disjointness Communication C
		Quasigroup Idempotent Density Inversely Proportional to ACC ⁰ Circ

Time	Verdict	Title
2026-05-12 17:37 UTC	INCONCLUSIVE	Free Cumulant Sum Gap in Read-Twice BPs for IP ₂
2026-05-12 17:56 UTC	INCONCLUSIVE	Tropical Variety Dimension Gap in Read-Twice vs Read-Once BPs
2026-05-12 18:08 UTC	INCONCLUSIVE	Real Rank of SOS Moment Matrices for Max-CUT Bounded by Degree
2026-05-12 18:22 UTC	INCONCLUSIVE	Submodular Rank Gap in Monotone DNF for k-CLIQUE

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```